

3.2 What factors need to be considered when developing design solutions for manufacture?

- a. Awareness of the responsibilities and principles of designing for manufacture (DFM), including:
 - i. planning for accuracy and efficiency through testing and prototyping
 - ii. being aware of issues in relation to different scales of production
 - iii. designing for repair and maintenance
 - iv. designing with consideration of product life.

- | | |
|----|--|
| b. | Awareness of product lifecycles that extend useful product life through planning for and consideration of maintenance, repair, upgrades, remanufacture and recycling systems. |
| c. | <p>Demonstrate an understanding of how environmental factors impact on:</p> <ul style="list-style-type: none"> i. sourcing and processing raw materials into a workable form ii. the disposal of waste, surplus materials and components, by-products of production including pollution related to energy iii. cost implications related to materials and process. |
| d. | <p>Demonstrate an understanding of sustainability issues relating to industrial manufacture, including:</p> <ul style="list-style-type: none"> i. fair trade and the Ethical Trade Initiative (ETI) ii. economic issues and globalisation iii. material sustainability and optimisation, availability, recycling and conservation schemes, such as: <ul style="list-style-type: none"> ○ exploring the impact and use of eco-materials ○ exploring how materials can be up-cycled. |